Unit circle

1

a $\frac{1}{6}$ b $\frac{1}{2}$ c $\frac{5}{4}$

2

a Quadrant 4 b Quadrant 1 c Quadrant 2 d Quadrant 3

e Quadrant 1 f Quadrant 4 g Quadrant 3 h Quadrant 2

3 0° or $360^\circ, 90^\circ, 180^\circ, 270^\circ$

4

a

i Quadrant 1 and Quadrant 2 ii Quadrant 3 and Quadrant 4

b

i Quadrant 1 and Quadrant 4 ii Quadrant 2 and Quadrant 3

c

i Quadrant 1 and Quadrant 3 ii Quadrant 2 and Quadrant 4

5

a Quadrant 2 b Quadrant 2 c Quadrant 2 d Quadrant 1

6

a Positive b Positive c Negative d Negative

e Negative f Negative g Positive h Negative

i Negative j Negative k Positive l Positive

m Positive n Positive o Negative p Positive

7

a

i $\frac{4}{5}$ ii $\frac{3}{5}$ iii $\frac{4}{3}$

b

i $\frac{20}{29}$ ii $-\frac{21}{29}$ iii $-\frac{20}{21}$

c

i $-\frac{21}{29}$ ii $\frac{20}{29}$ iii $-\frac{21}{20}$

8

a Yes b Yes c No d Yes

9

a False b False c True d True

e True

10

a True

b False

c True

d True

e False

11

a $2\sqrt{13}$

b

i $\frac{-3\sqrt{13}}{13}$

ii $\frac{-2\sqrt{13}}{13}$

iii $\frac{3}{2}$

12

a $\sin\theta = -\frac{4}{5}$

b $\tan\theta = -\frac{4}{3}$

13

a $\cos\theta = \pm\frac{3}{4}$

b $\tan\theta = \pm\frac{\sqrt{7}}{3}$

14

a $\cos\theta = -\frac{4}{5}$

b $\tan\theta = -\frac{3}{4}$

15

a $\cos\theta = \frac{10\sqrt{269}}{269}$

b $\sin\theta = -\frac{13\sqrt{269}}{269}$

16 $\cos\theta = -\frac{8}{17}$

17

a $\sin\theta = \frac{11}{61}$

b $\sin\theta = \frac{\sqrt{13}}{7}$

18

a $\tan\theta = \frac{2\sqrt{10}}{3}$

b $\tan\theta = \frac{1}{3}$

Equivalent ratios

19

20

a Quadrant 2 | b Negative | c 30° | d $-\cos 30^\circ$

21

a Quadrant 4 | b Negative | c 30° | d $-\tan 30^\circ$

22

a $\sin 87^\circ$ | b $-\cos 15^\circ$ | c $-\tan 61^\circ$ | d $-\sin 60^\circ$

e $\sin 33^\circ$ | f $-\tan 50^\circ$ | g $\sin 40^\circ$ | h $-\cos 65^\circ$

i $-\tan 15^\circ$ | j $-\sin 55^\circ$ | k $\cos 62^\circ$ | l $-\tan 2^\circ$

m $-\sin 41^\circ$ | n $-\cos 42^\circ$ | o $\tan 71^\circ$ | p $\sin 41^\circ$

q $-\cos 77^\circ$ | r $-\tan 40^\circ$

23

a $-\frac{b}{a}$ | b a | c b | d $-a$

e $-a$ | f $-b$

24

a

i $Q(-a, b)$ | ii $R(-a, -b)$ | iii $S(a, -b)$

b

i $\sin 49^\circ$ | ii $-\sin 49^\circ$ | iii $-\sin 49^\circ$

c

i $\sin x$ | ii $-\sin x$ | iii $-\sin x$

25

a

i $Q(-a, b)$ | ii $R(-a, -b)$ | iii $S(a, -b)$

b

i $-\cos 63^\circ$ | ii $-\cos 63^\circ$ | iii $\cos 63^\circ$

c

i $-\cos x$ | ii $-\cos x$ | iii $\cos x$

26

a

i $Q(-a, b)$ | ii $R(-a, -b)$ | iii $S(a, -b)$

b

i $-\tan 62^\circ$ | ii $\tan 62^\circ$ | iii $-\tan 62^\circ$

c

i $-\tan x$ ii $\tan x$ iii $-\tan x$

27

a $\sin\theta$ b $\cos\theta$ c $\sin\theta$ d $-\cos\theta$

e $-\sin\theta$ f $\cos\theta$ g $-\sin\theta$ h $\cos\theta$

28

a 0.93 b 0.36 c -0.93 d 0.36

e -0.36 f -0.93 g 0.36 h -0.93

Use technology

29

a 0.56 b -0.59 c -0.34 d 0.49

e -0.16 f -0.79 g -0.68 h 57.29

i 0.33 j 0.33 k 0.57 l -0.33

30

a 1 b 1 c 1 d 1

31

a $(0.819, 0.574)$ b 73° c $(-0.602, 0.799)$

32

a $(0.41, 0.91)$ b $(-0.41, 0.91)$ c 114° d $(0.41, -0.91)$

e 294°